Molecular Analysis of the Centrosomal Protein Cep152 in Mammalian Cells

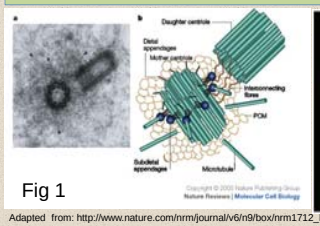
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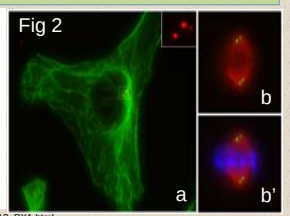
Introduction

- Discovered in late 19th century, the centrosome was described as "especial organ of cell division," because of its important contribution during cell division.
- The centrosome is present in most animal cells as an organelle serving as a primary microtubule organizing center (MTOC) essential for assembly of mitotic spindles in dividing cells.
- In spite of the long history of discovery and the important function in cells, molecular constituents of the centrosome and how they work together to orchestrate MTOC activities are not fully understood.
- With the recent advent of new technologies, many centrosome proteins have been identified, yet their localization and role in the centrosome remained elusive.
- In this study, we analyzed one recently identified centrosome protein termed Cep152 to identify its detailed localization at a particular subdomain of the centrosome and its possible involvement in microtubule organization in mammalian cells.

Background

- Centrosome is a multi-protein complex organelle composed of a pair of orthogonally arranged centrioles, mother and daughter, which is surrounded by an amorphous cloud of proteins called pericentriolar matrix (Fig. 1a). Each centriole is in a cylindrical shape consisted of a cartwheel of nine pairs of microtubule triplets (Fig. 1b) [1].
- Some centriolar proteins are distributed asymmetrically between two centrioles: centrobin is specifically localized to the daughter centriole as well as the procentrioles, while Cby is present at the distal end of the mother centriole.
- By serving as MTOCs in cells, the centrosome plays important biological functions, including mitosis. It locates at each spindle pole and assembles mitotic microtubules to form the spindle structure, which is required for equi-distribution of duplicated chromosomes into two daughter cells [3].
- Studies of the MTOC activity of centrosomes would be greatly facilitated by analysis of ectopic MTOCs induced in cells by overexpression of one of the centrosomal proteins, centrosomin (CNN) originally identified in *Drosophila melanogaster* [4].

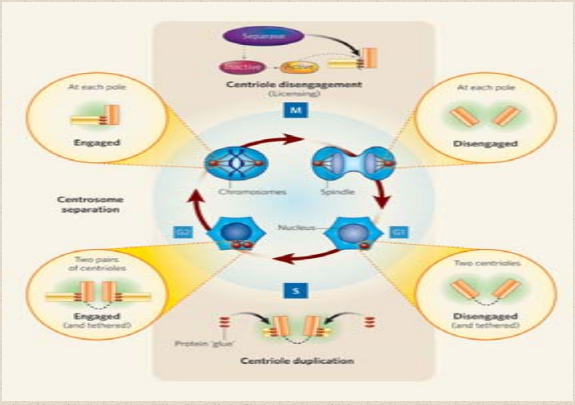




Adapted from: http://www.nature.com/nrm/journal/v6/n9/box/nrm1712_BX1.html

Fig 1. Mammalian centrosome Electron micrograph (a) and a schematic diagram of the centrosome (b). **Fig 2. Immunofluorescence micrographs.** Interphase (a) and mitotic (b-b') CHO (Chinese hamster ovary) cells were stained with Cep152, microtubules, and nucleus/chromosomes. Cep152 is seen at the interphase centrosomes and mitotic spindle poles.

Centrosome Cycle



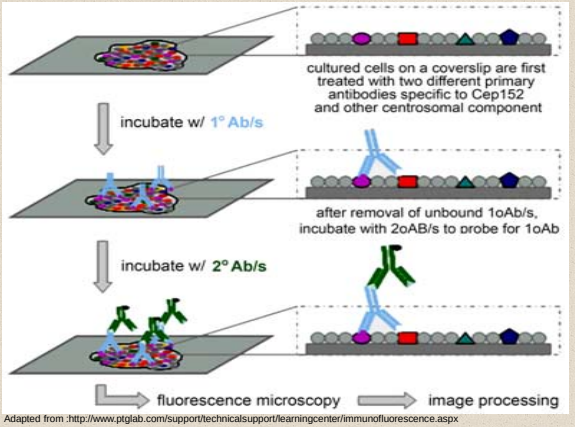
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Fig 3. Centrosome cycle. Arrangement of the centrioles is shown at different cell cycle stages (G1, S, G2, M). Each centrosome indicated by a circle consists of different centrioles (mother, daughter, and procentrioles; orange/yellow cylinders) surrounded by the pericentriolar matrix (green).

Significance of the Study

There is increasing evidence suggesting that the centrosome is involved in a wide range of human diseases, such as microcephaly and ciliopathies. Cancer cells frequently induce centrosomes abnormal in number, MTOC activity, and macromolecular compositions. Through the in-depth understanding of the molecular mechanism of centrosomal structure and function, we aspire to develop effective strategies of prevention, diagnosis and intervention against these serious human diseases [2].

Method



Adapted from :http://www.ptglab.com/support/technicalsupport/learningcenter/immunofluorescence.aspx

Fig 4. Immunostaining Technique

Results

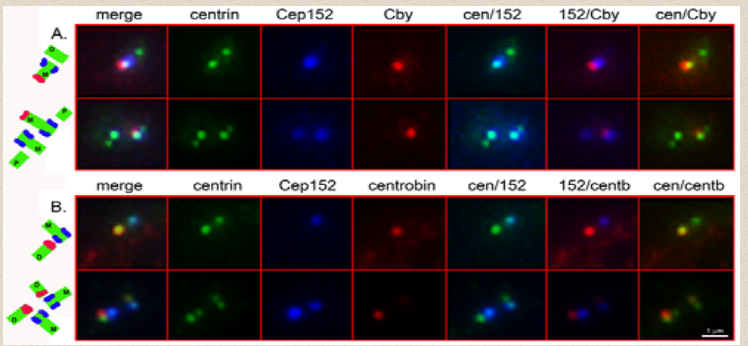


Fig 5. Localization of Cep152 at the mother centriole. CHO cells expressing GFP-tagged centriolar protein, centrin (green), were immunostainined with antibodies to Cep152 (blue) and other centriolar proteins (red), either mother centriole-specific Cby or daughter centrioles-specific centrolobin. M: mother centriole, D: daughter centriole, P: procentriole

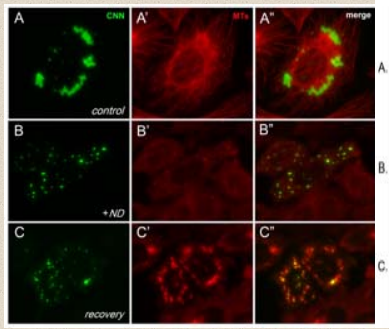


Fig 6. Centrosomine (CNN) makes ectopic MTOC. A-A" Control cells expressing CNN (green) capable of nucleating microtubules (MTs) (red). Merged image is shown in (A"). **B-B"** Cells were treated with nocodazole to depolymerize microtubules. **C-C"** After washing out nocodazole, short microtubules were formed onto the CNN aggregates.

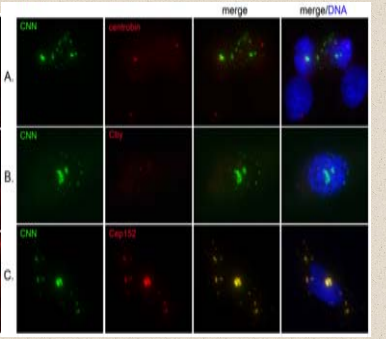


Fig 7. Colocalization of Cep 152 with centrosomin (CNN). **A** Centrobin (red) is partially co-localized with CNN (green). **B** Cby (red) partial overlaps with CNN. **C** Cep152 (blue) is perfectly aligned with CNN, suggesting that Cep152 interacts with CNN, directly or indirectly.

Discussion

- The results indicate that Cep152 is preferentially localized to the mother centrioles.
- Unlike other centrosomal proteins, such as Cby and centrobin, Cep152 is specifically co-localized at the ectopic MTOCs in cells overexpressing CNN. These results strongly indicate that Cep152 is directly involved in the formation of MTOC structures.
- In future, we will further analyze the effect of up and down regulation of Cep152 on microtubule organization using overexpression vectors and siRNA transfection strategies respectively.

References

1. Bornes, M. (2002) Centrosome composition and microtubule anchoring mechanisms. *Curr. Opin. Cell Biol.* 14:25-34.
2. Nigg, E.A. (2002) Centrosome aberrations: cause or consequence of cancer progression? *Nat. Rev. Cancer* 2:815-825.
3. Nigg, E.A. & Stearns, T. (2011) The centrosome cycle: Centriole biogenesis, duplication and inherent asymmetries. *Nat. Cell Biol.*13:1154-1160.
4. Terada, Y., Uetake, Y. & Kuriyama, R. (2003) Interaction of Aurora-A and centrosomin at the microtubule-nucleating site in Drosophila and mammalian cells. *J. Cell Biol.* 162:757-753.